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$f(x, y, z) = xyz$ in $a(1, 1, 1)$ in de richting $h\left(\frac{1}{2}, 0, \frac{1}{2}\right)$

$$f(1, 1, 1) = 1$$

$$Df(a, h) = \lim_{\lambda \rightarrow 0} \frac{f\left(1 + \frac{1}{2}\lambda, 1, 1 + \frac{1}{2}\lambda\right) - f(1, 1, 1)}{\lambda}$$

$$= \lim_{\lambda \rightarrow 0} \frac{\left(1 + \frac{1}{2}\lambda\right)^2 - 1}{\lambda}$$

$$= \lim_{\lambda \rightarrow 0} \frac{1 + \lambda + \frac{1}{4}\lambda^2 - 1}{\lambda}$$

$$= \lim_{\lambda \rightarrow 0} \left(1 + \frac{1}{4}\lambda\right) = 1$$

References